

(TR-104) PYTHON WITH AI/ML

Training Day 88 Report

Date: 13 May 2026

Topic Covered: Backend Security Enhancements, Query Stability, Face Recognition Authentication & QR Workflow Analysis

The eighty-eighth day of training focused on improving backend security and frontend connectivity in the MB AI Software project, enhancing database query stability, analyzing face recognition-based authentication workflows, and reviewing QR code-based classroom joining mechanisms.

- Configured CORS middleware in the FastAPI backend to enable secure communication between frontend and backend systems.
- Resolved frontend connectivity issues and verified successful API integration and data exchange.
- Improved MongoDB query handling for inputs containing special characters.
- Implemented regex sanitization techniques to prevent query failures and improve backend stability.
- Enhanced search and filtering functionalities across customer, trailer, payment, and insurance modules.
- Performed testing and validation of database retrieval functionalities after backend updates.
- Tested multiple query variations and typo-based inputs to verify system reliability and error handling.
- Studied the face recognition pipeline used in the college project.
- Analyzed Dlib-based face detection, embedding generation, and SVM classifier workflows.
- Reviewed attendance prediction mechanisms and face embedding comparison techniques.
- Worked on student authentication workflows using FaceID-based login functionality.
- Integrated camera-based face capture and recognition processes for user authentication.

- Implemented registration workflows for new users and analyzed classifier retraining mechanisms.
- Reviewed session management and login state handling after successful authentication.
- Analyzed QR code-based subject joining functionality and reviewed the QR generation workflow.
- Studied the integration approach used for classroom access and sharing mechanisms.
- Revised concepts related to CORS configuration, regex sanitization, MongoDB query optimization, face recognition systems, Dlib, SVM classifiers, authentication workflows, and QR code integration.

Overall, the training enhanced understanding of backend security, database query reliability, facial recognition-based authentication systems, QR code workflows, and frontend-backend integration techniques.